Quantum Gas Transport

Numerical method, verification and applications

A C++17 Galerkin solver for the linearized Uehling–Uhlenbeck equation

Luigi Di Gregorio

Technical note — version 0.4.1
10 October 2026

Abstract

This note describes a reproducible C++17 implementation of transport and linear relaxation for normal classical, Bose and Fermi gases. Matched Sonine–Laguerre bases, rotational reduction of collision integrals from eight to five variables and Gaussian quadrature yield finite positive-definite systems for viscosity and thermal conductivity. Five first-profile states agree with a separately evaluated variational reference within 0.0031%; basis and quadrature refinements assess the separate discretization errors. The same collision matrices also generate homogeneous linear relaxation and a projected transverse-wave calculation. An optional final application couples the computed Bose transport coefficients to normal-phase thermodynamics to compare sound damping with published measurements. A classical-transport control with identical thermodynamics yields nearly the same standardized RMS residuals on the five selected observations; the comparison does not establish improved agreement from quantum transport corrections. The implementation is in C++17. The scope is deterministic linear kinetic theory, with a separate assessment of the additional assumptions in the sound application.

Contents

1 Model and numerical transport solver **2**
1.1 Kinetic model and transport problem . . . 2
1.2 Normalization: separating physics from units . . . 2
1.3 The collision kernel used for the first comparison . . . 3
1.4 A quantum basis that satisfies matching . . . 4
1.5 From the integral equation to a matrix problem . . . 5
1.6 How the collision integrals are evaluated . . . 5
1.7 Verification and convergence are different questions . . . 6
1.8 Transport results at equal approximation order . . . 6
1.9 Implementation and reproducibility . . . 7

2 Using the collision matrices to simulate linear relaxation **8**
2.1 A time-dependent question with the same microscopic model . . . 8
2.2 Generalized eigenvalues and the entropy check . . . 9
2.3 The initial perturbation and the observable . . . 9
2.4 A transverse wave: recovering viscosity from dynamics . . . 11

3 Optional application: normal-phase sound damping **12**
3.1 What is compared with the experiment? . . . 12
3.2 An experimentally motivated kernel and units . . . 12
3.3 Why the thermodynamic model must also change . . . 13
3.4 Deriving the sound eigenproblem . . . 13
3.5 Results and the limits of this comparison . . . 14

1 Model and numerical transport solver

1.1 Kinetic model and transport problem

Let $f(t, \mathbf{x}, \boldsymbol{\xi})$ be the number density in position–velocity space, with $n = \int f \, \mathrm{d}^3\xi$ and $\mathbf{u} = n^{-1} \int \boldsymbol{\xi} f \, \mathrm{d}^3\xi$. For normal, nonrelativistic gases with elastic binary collisions, the Uehling–Uhlenbeck model [1] is

$$\begin{aligned} \partial_t f + \boldsymbol{\xi} \cdot \nabla f &= Q_\varepsilon(f), \\ Q_\varepsilon(f) &= \int B \left[f' f_1' (1 + \varepsilon A f)(1 + \varepsilon A f_1) - f f_1 (1 + \varepsilon A f')(1 + \varepsilon A f_1') \right] \mathrm{d}^3\xi_1 \, \mathrm{d}\Omega. \end{aligned} \tag{1}$$

Primes denote post-collision velocities; subscript 1 labels the second particle. The signs $\varepsilon = +1, -1, 0$ describe Bose, Fermi and classical statistics. The constant $A = h_\mathrm{P}^3/(g_s m^3)$ converts number density in velocity space to occupation per internal state, where g_s is the internal degeneracy and h_P is Planck's constant. The collision measure is nonnegative and invariant under particle exchange and time reversal. We assume isotropic scattering and sufficient angular support to retain only the usual five collision invariants.

With $\mathbf{U} = \boldsymbol{\xi} - \mathbf{u}$, a regular local equilibrium is

$$f^{(0)} = \frac{1}{A} \frac{1}{z^{-1} \exp[mU^2/(2k_\mathrm{B}T)] - \varepsilon}, \qquad z = \exp[\mu_\mathrm{chem}/(k_\mathrm{B}T)]. \tag{2}$$

The classical form is understood as the $\varepsilon = 0$ convention. Collision balance holds at local equilibrium, while spatial gradients generate a small correction. At first Chapman–Enskog order, rotational symmetry reduces that correction to two unknown radial functions:

$$\delta f = w\chi, \qquad \chi = -a(U) U_i \partial_i \ln T - b(U) D_{ij}(\mathbf{U}) \mathring{S}_{ij}, \qquad D_{ij} = U_i U_j - \tfrac{1}{3} U^2 \delta_{ij}. \tag{3}$$

Here $w = f^{(0)}(1 + A\varepsilon f^{(0)})$ and $\mathring{S}_{ij} = (\partial_i u_j + \partial_j u_i)/2 - \delta_{ij}(\nabla \cdot \mathbf{u})/3$ is the traceless strain rate. Repeated Cartesian indices are summed. The functions a and b encode the response to a temperature gradient and a shear, respectively. Symmetry determines their angular structure, but it does not determine their radial dependence or their size. The kinetic stress and heat flux are

$$p_{ij} = m \int U_i U_j f \, \mathrm{d}^3\xi, \qquad q_i = \frac{m}{2} \int U^2 U_i f \, \mathrm{d}^3\xi. \tag{4}$$

Their first-order constitutive forms define the two coefficients: $p_{ij} = p\delta_{ij} - 2\mu \mathring{S}_{ij}$ and $q_i = -\lambda \partial_i T$. At equilibrium, odd heat-flux moments vanish and stress is isotropic; the correction supplies these dissipative fluxes.

Our numerical problem is therefore concrete: choose an equilibrium state and a collision kernel, solve the two linear collision equations approximately, and evaluate the moments that give λ and μ. These coefficients are outputs, not fitted parameters. The calculation applies to dilute normal gases with elastic binary collisions, close to local equilibrium. It does not include a condensate, a superfluid component or a strongly interacting liquid.

Galerkin projection, Sonine–Laguerre expansions and Gaussian quadrature are established numerical tools; the contribution is their implementation, verification and application to the specified model. We do not implement Wu's fast Fourier collision algorithm. We use a reference formula from that work as one verification route [2].

1.2 Normalization: separating physics from units

For particle mass m, the mass density is $\rho = mn$. We introduce

$$v_0 = \sqrt{\frac{k_\mathrm{B}T}{m}}, \qquad \mathbf{v} = \frac{\mathbf{U}}{v_0}, \qquad E = \frac{v^2}{2}, \qquad \widehat{f} = \frac{v_0^3}{n} f. \tag{5}$$

Thus $\int \widehat{f}\,\mathrm{d}^3v = 1$. The normalization is conditional on the local density: the physical f integrates to n, while $\widehat{f}$ integrates to one. All polynomial arguments below use the dimensionless energy E.

Define the statistical integrals by

$$G_\nu^{(\varepsilon)}(z) = \frac{1}{\Gamma(\nu)} \int_0^\infty \frac{E^{\nu-1}}{z^{-1}e^E - \varepsilon}\,\mathrm{d}E. \tag{6}$$

For $\varepsilon = 0$, $G_\nu = z$. The integral definition also works for Fermi fugacities greater than one, where a small-z power series is unsuitable. Writing

$$D(E) = 1 - \varepsilon z e^{-E}, \qquad \mathcal{N} = (2\pi)^{3/2}\frac{G_{3/2}^{(\varepsilon)}(z)}{z}, \tag{7}$$

the normalized equilibrium and its linearization weight are

$$\widehat{f}_0 = \frac{e^{-E}}{\mathcal{N}D(E)}, \qquad \widehat{A} = z\mathcal{N}, \qquad r = 1 + \varepsilon\widehat{A}\widehat{f}_0 = \frac{1}{D(E)}, \qquad w = \widehat{f}_0 r. \tag{8}$$

In the rest of this numerical section w, the profiles and the collision operator are dimensionless. The pressure is

$$Z_{\mathrm{kin}} = \frac{p}{nk_{\mathrm{B}}T} = \frac{G_{5/2}^{(\varepsilon)}(z)}{G_{3/2}^{(\varepsilon)}(z)}. \tag{9}$$

We use $0 < z < 1$ for normal Bose equilibria and $z > 0$ for Fermi equilibria. The classical limit is recovered as occupations become small. Equal fugacity for two different statistics does not in general mean equal physical density at fixed temperature: $n\lambda_{\mathrm{dB}}^3$ also depends on $G_{3/2}$ and on the internal-state degeneracy. The dimensionless comparisons below isolate the transport calculation under the stated normalization.

1.3 The collision kernel used for the first comparison

Our baseline uses an isotropic hard-sphere-type differential cross section,

$$B(g,\Omega) = g\frac{d^2}{4}, \qquad g = |\mathbf{U} - \mathbf{U}_1|, \qquad \sigma_{\mathrm{tot}} = \int_{S^2}\frac{d^2}{4}\,\mathrm{d}\Omega = \pi d^2. \tag{10}$$

Here d is the hard-sphere diameter; the numerical convention sets $d = 1$. The kernel includes both the relative speed, which controls encounter rates, and the differential cross section, which describes scattering into a solid angle. Keeping exactly the same kernel when changing statistics isolates the effect of the occupation factors. This is a model comparison, not a claim that this cross section describes every atomic Bose or Fermi gas. In particular, real fermionic scattering depends on spin composition and allowed partial waves.

The implementation also permits isotropic power laws $B = Cg^\gamma$, $C > 0$, $\gamma > -3$. The radial quadrature weight changes with γ; the angular symmetry remains available. An arbitrary anisotropic scattering law would require a further extension.

For the baseline convention the physical units can be restored using

$$t_0 = \frac{1}{nd^2v_0}, \qquad \ell_0 = \frac{1}{nd^2}, \qquad \mu_{\mathrm{phys}} = \mu_{\mathrm{num}}\frac{\sqrt{mk_{\mathrm{B}}T}}{d^2}, \qquad \lambda_{\mathrm{phys}} = \lambda_{\mathrm{num}}\frac{k_{\mathrm{B}}\sqrt{k_{\mathrm{B}}T/m}}{d^2}. \tag{11}$$

The scale t_0 is our unit of time; it is not an independently calculated mean collision time, which also contains statistical and geometric factors.

1.4 A quantum basis that satisfies matching

With the equilibrium parameters held fixed, define the linearized operator by

$$\mathcal{L}\chi = \frac{\mathrm{d}}{\mathrm{d}s} Q_\varepsilon(f_0 + sw\chi)\Big|_{s=0}, \tag{12}$$

using the dimensionless variables and kernel in the numerical calculation. The linearized equations for one representative vector component and one off-diagonal tensor component are

$$\mathcal{L}A_x = -wJ_x, \qquad J_x = v_x(E - \tfrac{5}{2}Z_{\mathrm{kin}}), \qquad \mathcal{L}B_{xy} = -wv_xv_y, \tag{13}$$

where $A_x = a(v)v_x$ and $B_{xy} = b(v)v_xv_y$. There is no sum over x or y here. The tensor component suffices because rotational symmetry makes every shear orientation equivalent. For a flow $\partial_y u_x = G$, the two terms $\mathring{S}_{xy} = \mathring{S}_{yx} = G/2$ combine to give $b(v)v_xv_yG$, so no extra factor of two is missing.

We approximate the radial functions by N profiles,

$$a_N(v) = \sum_{n=1}^{N} \alpha_n P_n(E), \qquad b_N(v) = \sum_{n=1}^{N} \beta_n Q_n(E). \tag{14}$$

For viscosity we take

$$Q_n(E) = L_{n-1}^{(5/2)}(E). \tag{15}$$

For heat conduction we explicitly remove the momentum component:

$$\boxed{P_n(E) = L_n^{(3/2)}(E) - s_n, \qquad s_n = \frac{\int wv^2 L_n^{(3/2)}(E)\,\mathrm{d}^3v}{\int wv^2\,\mathrm{d}^3v}.} \tag{16}$$

Indeed, substituting this definition gives $\int wv^2 P_n\,\mathrm{d}^3v = 0$. Therefore every finite linear combination a_N satisfies thermal momentum matching, regardless of the coefficients α_n. Thermal mass and energy matching follow from oddness. In the viscous sector the tensor structure ensures all matching conditions by parity and tracelessness.

For reference, the generalized Laguerre polynomials are generated by

$$L_0^{(\alpha)}(E) = 1, \qquad L_1^{(\alpha)}(E) = \alpha + 1 - E, \tag{17}$$

$$(n+1)L_{n+1}^{(\alpha)}(E) = (2n + \alpha + 1 - E)L_n^{(\alpha)}(E) - (n+\alpha)L_{n-1}^{(\alpha)}(E). \tag{18}$$

In the classical case, $s_n = 0$ for $n \geq 1$ by Laguerre orthogonality. In the quantum case the shifts are generally nonzero. For example, the weighted moment identities yield

$$s_1 = \frac{5}{2}(1 - Z_{\mathrm{kin}}), \qquad P_1(E) = \frac{5}{2}Z_{\mathrm{kin}} - E. \tag{19}$$

Thus the first thermal vector is $P_1 v_x = -J_x$. This is a useful explicit example of how matching changes a classical polynomial in a quantum gas.

These shifted polynomials need not be mutually orthogonal under the quantum weight. They only need to span an admissible, linearly independent trial space. Galerkin projection handles their couplings through a full matrix. Orthogonality of the classical polynomials would not diagonalize the collision operator either. Increasing N improves radial resolution; it does not increase the Chapman–Enskog order, which remains first order in spatial gradients.

1.5 From the integral equation to a matrix problem

Define $\Phi_n^T = P_n(E)v_x$ and $\Phi_n^S = Q_n(E)v_xv_y$, and use the bracket

$$[\phi,\psi] = -\int \phi\mathcal{L}\psi\,\mathrm{d}^3v = \frac{1}{4}\int BW\,\Delta\phi\,\Delta\psi\,\mathrm{d}^3v\,\mathrm{d}^3v_1\,\mathrm{d}\Omega, \quad \Delta\phi = \phi+\phi_1-\phi'-\phi_1'. \tag{20}$$

Here $W = \widehat{f}_0\widehat{f}_{0,1}r'r_1'$ is the detailed-balance weight of the entire collision quadruple. It is distinct from the one-velocity weight w in the matching conditions.

For example, the thermal residual is $R_N = \mathcal{L}(\sum_n \alpha_n\Phi_n^T) + wJ_x$. Imposing $\int \Phi_m^T R_N\,\mathrm{d}^3v = 0$ for each m gives

$$\boxed{M^T\boldsymbol{\alpha} = \boldsymbol{d}^T, \qquad M_{mn}^T = [\Phi_m^T,\Phi_n^T], \qquad d_m^T = \int w\Phi_m^T J_x\,\mathrm{d}^3v.} \tag{21}$$

The superscript T labels the thermal sector here, not a matrix transpose. Similarly,

$$M^S\boldsymbol{\beta} = \boldsymbol{d}^S, \qquad M_{mn}^S = [\Phi_m^S,\Phi_n^S], \qquad d_m^S = \int w\Phi_m^S v_xv_y\,\mathrm{d}^3v. \tag{22}$$

The bracket is nonnegative. Once the collision-invariant kernel is removed, an admissible independent finite basis gives a positive-definite matrix under the collision assumptions stated above. We solve these systems by Cholesky factorization and check symmetry, positivity and residuals numerically.

The transport coefficients can then be evaluated in two equivalent ways:

$$\boxed{\lambda_N = \sum_m d_m^T\alpha_m = \frac{1}{6}\int wa_N(v)v^4\,\mathrm{d}^3v, \qquad \mu_N = \sum_m d_m^S\beta_m = \frac{1}{15}\int wb_N(v)v^4\,\mathrm{d}^3v.} \tag{23}$$

For the first equality for heat conduction, isotropy gives $\int wa_Nv_x^2(E-5Z_{\mathrm{kin}}/2) = \frac{1}{3}\int wa_Nv^2(E-5Z_{\mathrm{kin}}/2)$. Matching removes the term proportional to v^2, and $E = v^2/2$ gives $1/6$. For viscosity, $\int R(v)v_x^2v_y^2 = \frac{1}{15}\int R(v)v^4$ for $x \neq y$. This also explains why solving for a single component introduces no missing factor of three.

For either sector, $d^{\mathsf{T}}M^{-1}d$ is the maximum over trial coefficients c of $2d^{\mathsf{T}}c - c^{\mathsf{T}}Mc$. Nested trial spaces therefore give nondecreasing transport estimates when the integrals are evaluated consistently and exactly. A small increment with N is useful evidence of convergence, but it is not a rigorous error bound for the infinite-dimensional solution.

1.6 How the collision integrals are evaluated

A direct bracket has eight integration variables: three for each incoming velocity and two for the outgoing direction. Introduce centre and relative velocities,

$$\mathbf{V} = \frac{\mathbf{v}+\mathbf{v}_1}{2}, \quad \mathbf{q} = \frac{\mathbf{v}-\mathbf{v}_1}{2}, \quad \mathbf{v} = \mathbf{V}+\mathbf{q}, \quad \mathbf{v}_1 = \mathbf{V}-\mathbf{q}. \tag{24}$$

Elastic scattering replaces $\mathbf{q}$ by $\mathbf{q}'$ with $q' = q$. The Jacobian is $\mathrm{d}^3v\,\mathrm{d}^3v_1 = 8\,\mathrm{d}^3V\,\mathrm{d}^3q$. We can analytically average over the three angles of a common rotation, because the equilibrium and kernel are isotropic.

Choose $\mathbf{V}$ along the polar axis and the incoming $\mathbf{q}$ in a fixed meridian plane. Five variables remain:

$$X = V^2, \quad Y = q^2, \quad a = \cos\theta_{\mathrm{in}}, \quad b = \cos\theta_{\mathrm{out}}, \quad \varphi \in [0,2\pi). \tag{25}$$

The first two range over $[0,\infty)$, and $a,b \in [-1,1]$. The incoming energies are $E = (X+Y+2\sqrt{XY}a)/2$ and $E_1 = (X+Y-2\sqrt{XY}a)/2$; replace a by b for the outgoing energies. The azimuth still enters the vector and tensor collision increments.

For $B = Cg^\gamma$, the reduced measure, including the factor $1/4$ in the bracket and the normalization of the equilibrium, is

$$4\,2^\gamma C\,X^{1/2}Y^{(1+\gamma)/2}e^{-X-Y}\mathcal{R}\,\mathrm{d}X\,\mathrm{d}Y\frac{\mathrm{d}a}{2}\frac{\mathrm{d}b}{2}\frac{\mathrm{d}\varphi}{2\pi}, \qquad \mathcal{R} = \frac{1}{(G_{3/2}/z)^2 DD_1D'D'_1}. \tag{26}$$

This formula shows precisely where the quantum modification is evaluated: at all four collision energies, rather than only at one incoming velocity. For the baseline $C = 1/4$, $\gamma = 1$, the numerical prefactor is 2.

For vector profiles the rotational average of two selected-component increments is $\Delta\mathbf{A}_m \cdot \Delta\mathbf{A}_n/3$. For traceless symmetric tensor profiles the average for a selected off-diagonal component is $\Delta\mathsf{B}_m : \Delta\mathsf{B}_n/10$. These identities retain the component conventions used in the sources.

The two radial integrals use generalized Gauss–Laguerre rules with weights $X^{1/2}e^{-X}$ and $Y^{(1+\gamma)/2}e^{-Y}$. The two polar cosines use Gauss–Legendre rules; the periodic azimuth uses a trapezoidal rule. Gaussian nodes and weights are constructed from the eigenproblem of the corresponding Jacobi matrix. No sharp velocity cutoff is needed for these radial integrals.

A typical transport run uses $(n_r, n_p, n_\varphi) = (24, 20, 24)$, namely $n_r^2 n_p^2 n_\varphi = 5{,}529{,}600$ collision configurations. Near the Bose threshold the factors $D(E)^{-1}$ vary more rapidly at low energy; the main scan raises n_r to 40 for $z \geq 0.8$. Separate refinement tests also increase angular orders. This cost is paid to assemble small $N \times N$ matrices; solving the linear systems is not the expensive step.

1.7 Verification and convergence are different questions

First-profile verification. For a classical hard-sphere gas, the one-profile values in our units are

$$\mu_1 = \frac{5}{16\sqrt{\pi}}, \qquad \lambda_1 = \frac{75}{64\sqrt{\pi}}. \tag{27}$$

The quantum one-profile result can also be evaluated through the separate integral representation in Eq. (50) of Wu's work [2], after matching normalizations. Specifically, the quantum comparison uses ratios to the classical one-profile values: constant scattering and spin multiplicities in the original reference cancel in these ratios. We implement that reference through a distinct, four-variable integration route. Across the classical case, Bose $z = 0.5, 0.9$ and Fermi $z = 1, 10$, the largest relative discrepancy in either coefficient is 3.03×10^{-5}, or less than 0.0031%. At $N = 1$, Galerkin reduces to $M_{11}\alpha_1 = d_1$ in each sector, and the corresponding dimensionless coefficient is d_1^2/M_{11}. Wu's variational ansatz has this same one-profile form; Eq. (50) expresses its result using reduced integrals I_A and I_B. We evaluate those integrals directly in a separate code path, rather than calling the bracket routine. The two paths share equilibrium helper functions. This checks the assembly against the same one-profile theory. It does not establish 0.0031% accuracy of a finite-N solution to the full kinetic equation, nor agreement with an experiment.

Basis and quadrature refinement. We separately vary N at fixed integration settings, and refine the quadrature at fixed $N = 4$. Table 1 reports the relative change from $N = 4$ to $N = 5$, and the change between quadratures $(24, 16, 20)$ and $(40, 24, 28)$. The latter uses 25,804,800 collision configurations. The largest displayed $N = 4 \to 5$ change is about 0.3134%, in Fermi conductivity at $z = 10$. The basis truncation can therefore matter more than the benchmark discrepancy. Figure 1 uses $N = 5$ as a common reference to show the trend, without treating it as an exact solution.

1.8 Transport results at equal approximation order

Figure 2 compares each quantum result to the classical result at the *same* N. This avoids mixing a fourth-profile quantum calculation with a first-profile classical denominator. The main

Statistics	z	$N = 4 \to 5$ change [%] μ	λ	Quadrature change [%] μ	λ
classical	–	0.00214	0.00613	0.00000	0.00000
bose	0.9	0.02260	0.27886	0.00355	0.00310
fermi	10.0	0.16354	0.31339	0.00000	0.00000

Table 1: Absolute relative changes, normalized to the coarser result. Values printed as zero are below the displayed precision. These are convergence diagnostics, not certified error bars.

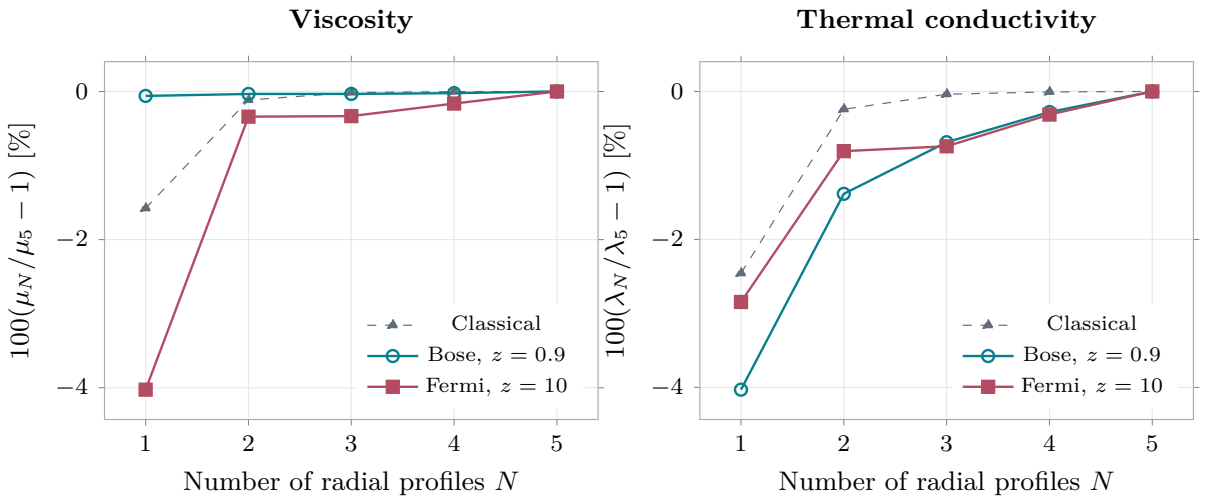


Figure 1: Basis refinement for the classical gas, Bose $z = 0.9$ and Fermi $z = 10$. Each curve is divided by its own $N = 5$ result. The vanishing endpoint is a consequence of this normalization, not proof of convergence at $N = 5$.

scan contains 36 transport calculations: 18 statistical states evaluated at $N = 1$ and $N = 4$. Representative $N = 4$ values are listed in Table 2. For this kernel and these states, Bose statistics

Statistics	z	μ_4	λ_4	$\mu_4/\mu_{\mathrm{cl},4}$	$\lambda_4/\lambda_{\mathrm{cl},4}$
classical	–	0.179131	0.677778	1.000000	1.000000
bose	0.5	0.142615	0.651371	0.796150	0.961040
bose	0.9	0.094207	0.620652	0.525913	0.915716
fermi	1.0	0.229766	0.714132	1.282666	1.053638
fermi	10.0	0.441023	0.853381	2.462011	1.259087

Table 2: Dimensionless coefficients and same-order ratios. Physical units are restored by Eq. (11).

lowers both coefficients relative to the classical values, whereas Fermi statistics raises them. The change in conductivity is appreciably smaller than the change in viscosity over this scan. These are computed model results. Occupation factors alone do not establish a universal monotonicity theorem for transport coefficients under arbitrary changes of cross section and thermodynamic constraints.

1.9 Implementation and reproducibility

The solver is written in C++17 with Eigen for linear algebra and a JSON library for structured results. It includes equilibrium moments, matched polynomials, Gaussian rules, collision matrices, transport solves and the dynamic extensions below. The numerical test executable reports 24 test groups, supplemented by command-line smoke tests in CTest. Build commands, local verification

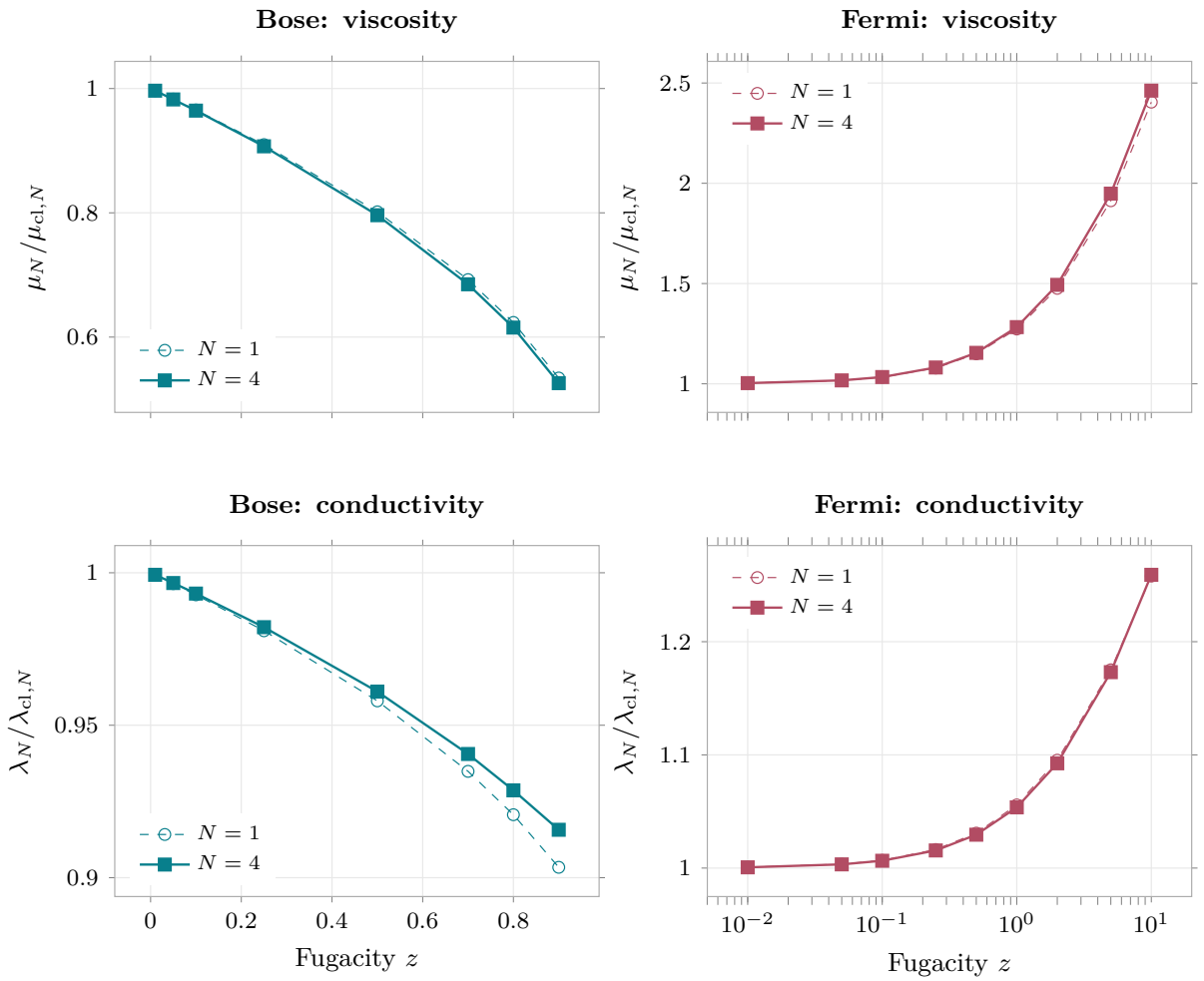


Figure 2: Quantum-to-classical transport ratios for the same isotropic kernel and the same number of radial profiles in numerator and denominator. Changing N changes the numerical approximation, while changing z changes the equilibrium occupation. All displayed Bose states are above the condensation threshold in the normal-gas description.

records and CI configuration accompany the source. These software checks complement the physical benchmarks and convergence studies.

The migration from the earlier Python implementation was also checked against saved results, retained as attributed regression fixtures. This is a cross-implementation consistency check, not an independent physical benchmark. Every figure in this note is generated from the saved C++ outputs, without fitting or manually adjusting the transport curves. The source files, calculation settings and tabulated outputs are kept alongside the plotting data.

2 Using the collision matrices to simulate linear relaxation

2.1 A time-dependent question with the same microscopic model

The transport problem asks how a persistent gradient drives a small departure from equilibrium. A complementary question is what happens if a departure is prepared and then allowed to relax, without any spatial gradient or external force. Around a fixed homogeneous equilibrium, the linearized equation is

$$w\partial_t\chi = \mathcal{L}\chi. \tag{28}$$

The weight w is time independent in this calculation. Matching has already removed the mass, momentum and energy modes: the perturbation changes a heat flux or a shear stress, not the equilibrium parameters.

In either of the two finite trial spaces, write $\chi_N(t,\mathbf{v}) = \sum_n c_n(t)\Phi_n(\mathbf{v})$ and test Eq. (28) against Φ_m. Define the Gram matrix

$$H_{mn} = \int w\Phi_m\Phi_n \,\mathrm{d}^3v. \tag{29}$$

The time derivative contributes $\sum_n H_{mn}\dot{c}_n$; the collision term contributes $-\sum_n M_{mn}c_n$. Therefore

$$\boxed{H\dot{\boldsymbol{c}} = -M\boldsymbol{c}.} \tag{30}$$

This is an actual evolution generated by the collision operator projected onto the chosen space. Its rates are not inserted as adjustable decay times.

2.2 Generalized eigenvalues and the entropy check

Since both M and H are positive definite in the matched finite space, solve the symmetric generalized eigenproblem

$$M\boldsymbol{v}_r = \nu_r H\boldsymbol{v}_r, \qquad \boldsymbol{v}_r^{\mathsf{T}} H\boldsymbol{v}_s = \delta_{rs}, \qquad \nu_r > 0. \tag{31}$$

The solution is

$$\boldsymbol{c}(t) = \sum_r a_r e^{-\nu_r t}\boldsymbol{v}_r, \qquad a_r = \boldsymbol{v}_r^{\mathsf{T}} H\boldsymbol{c}(0). \tag{32}$$

Thus a general perturbation is a superposition of decaying modes, rather than necessarily a single exponential. The slowest mode that is actually excited dominates sufficiently late times.

The quadratic relative-entropy perturbation is

$$\mathcal{E}_N(t) = \frac{1}{2}\int w\chi_N^2 \,\mathrm{d}^3v = \frac{1}{2}\boldsymbol{c}^{\mathsf{T}} H\boldsymbol{c}. \tag{33}$$

It is the second-order term of the quantum H functional around equilibrium, up to the convention for conserved additive terms. Differentiating and using Eq. (30),

$$\frac{\mathrm{d}\mathcal{E}_N}{\mathrm{d}t} = \boldsymbol{c}^{\mathsf{T}} H\dot{\boldsymbol{c}} = -\boldsymbol{c}^{\mathsf{T}} M\boldsymbol{c} \le 0. \tag{34}$$

This is the finite-dimensional counterpart of collision-bracket dissipation, Eq. (20). It gives a direct diagnostic for the time evolution. The positive eigenvalues demonstrate decay in this trial space; they do not prove a spectral gap for the full infinite-dimensional quantum operator.

2.3 The initial perturbation and the observable

For each sector we choose $\boldsymbol{c}(0) = H^{-1}\boldsymbol{d}$ and measure $J(t) = \boldsymbol{d}^{\mathsf{T}}\boldsymbol{c}(t)$. This is a specified, reproducible perturbation aligned with the projected transport source. The normalization of its amplitude is immaterial for the plotted ratios. For the thermal sector, matching makes the source moment equivalent to the heat-flux moment; for the shear sector it is the off-diagonal stress moment. With this choice $a_r = \boldsymbol{v}_r^{\mathsf{T}}\boldsymbol{d}$, so

$$J(t) = \sum_r a_r^2 e^{-\nu_r t}, \qquad \mathcal{E}_N(t) = \frac{1}{2}\sum_r a_r^2 e^{-2\nu_r t}. \tag{35}$$

In particular both quantities decay monotonically in these runs. Another initial perturbation need not make every individual flux monotone, even though the quadratic entropy still decreases.

Integration over time also gives an internal consistency identity,

$$\int_0^\infty J(t)\,\mathrm{d}t = \sum_r \frac{a_r^2}{\nu_r} = \boldsymbol{d}^{\mathsf{T}} M^{-1} \boldsymbol{d}. \tag{36}$$

The last expression is the corresponding static transport coefficient in our units. This ties stationary response to relaxation in the same discretized model; it is not an independent molecular-dynamics or Kubo calculation.

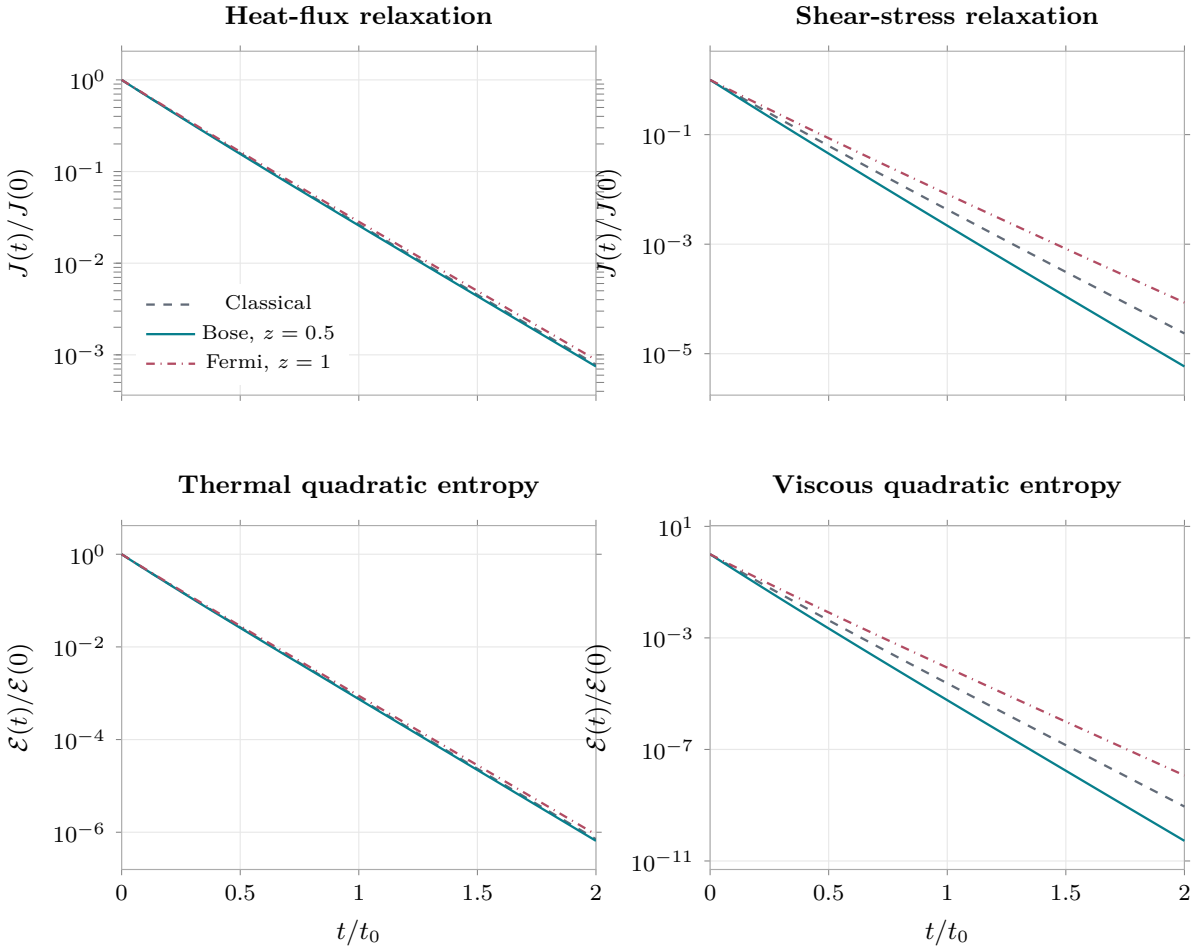


Figure 3: Homogeneous linear relaxation with $N = 4$ and the baseline kernel $B = g/4$. Time is measured in $t_0 = (nd^2v_0)^{-1}$. Upper panels show normalized source moments; lower panels show normalized quadratic relative entropy. The equilibrium is fixed, and the perturbation evolves according to $H\dot{c} = -Mc$.

Statistics	z	Smallest thermal $\nu_r t_0$	Smallest shear $\nu_r t_0$
classical	–	3.500646	5.054566
bose	0.5	3.519924	5.767084
fermi	1.0	3.445980	4.421469

Table 3: Smallest rates in the $N = 4$ matched trial spaces. These finite-space rates should not be identified with a proven full-operator gap.

Figure 3 is a deterministic simulation of a linearized distribution. It does not track particles, and it does not solve the nonlinear kinetic equation for large departures. Polynomial tangent per-

turbations also do not automatically ensure global positivity of a reconstructed finite-amplitude distribution at arbitrarily large velocity. The interpretation here is linear response.

2.4 A transverse wave: recovering viscosity from dynamics

We can also retain spatial transport. Consider a small flow in the x direction varying along y. A real spatial Fourier mode corresponds to $u_x(y,t) \propto \mathrm{Re}[\widehat{u}_x(t)e^{iky}]$. Its kinetic equation is

$$w(\partial_t\chi + v_y\partial_y\chi) = \mathcal{L}\chi. \tag{37}$$

In addition to the N thermal vectors and N shear tensors, include the conserved function v_x. It carries the mean flow and must not be removed when simulating momentum transport. The projected space has dimension $2N+1$. Define the streaming matrix

$$S_{mn} = \int w\Phi_m v_y \Phi_n \,\mathrm{d}^3v. \tag{38}$$

For the Fourier amplitudes the equation becomes

$$\boxed{H\dot{\boldsymbol{c}} = -(M + ikS)\boldsymbol{c}.} \tag{39}$$

We initialize the conserved velocity component and set the other coefficients to zero. Streaming then creates a shear stress, and collisions relax it. The coefficient μ is not supplied to this kinetic evolution.

At long wavelengths, the hydrodynamic prediction is

$$\partial_t u_x = \frac{\mu}{\rho}\partial_y^2 u_x, \qquad \widehat{u}_x(t) \simeq \widehat{u}_x(0)e^{-(\mu/\rho)k^2 t}. \tag{40}$$

If $s(k)$ is the slow eigenvalue of $-H^{-1}(M + ikS)$, the dynamic estimate is

$$\mu_{\mathrm{dyn}}(k) = -\rho\frac{\mathrm{Re}\, s(k)}{k^2}. \tag{41}$$

Agreement with the static coefficient as $k \to 0$ checks the coupling between streaming, collisions and the stress moment. It is a consistency test within the same projected model, rather than a second independent collision solver.

Statistics	μ_4	$\mu_{\mathrm{dyn}}(0.05)$	Relative difference [%]
classical	0.179131	0.179140	0.005002
bose	0.142615	0.142619	0.002269
fermi	0.229766	0.229786	0.009022

Table 4: Static and dynamic viscosities for the three states in Fig. 4. The largest listed difference is below 0.01%.

At finite k, streaming couples to angular sectors absent from this trial space. The displayed finite-wave-number evolution is consequently a reduced kinetic model. Resolving the full kinetic crossover would require convergence in those additional sectors, not only additional radial polynomials.

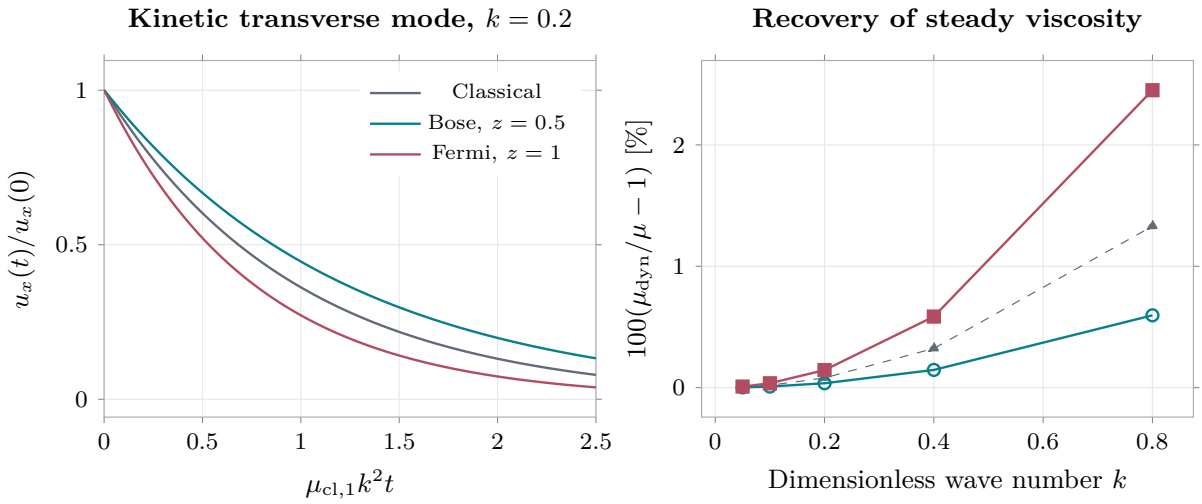


Figure 4: Left: kinetic transverse-mode amplitude at $k = 0.2$, compared with the Navier–Stokes exponential (dotted). A common classical time scale is used for all curves. Right: viscosity inferred from the slow eigenvalue approaches the static Galerkin value as k decreases. This transverse mode diffuses; it is not the oscillatory sound mode of the following section.

3 Optional application: normal-phase sound damping

3.1 What is compared with the experiment?

An experimental application requires more than changing the statistical sign. We need the particles' scattering cross section, their thermodynamic state and a model for the measured observable. We use the normal-phase sound measurements in Fig. 4 of Hilker et al. [3], with the public data in Ref. [4]. We do not model their superfluid second sound or claim to reproduce the complete experiment.

The chain of calculations is

$$\text{scattering model} \longrightarrow (\mu, \lambda)_{\text{Galerkin}} \longrightarrow \text{linear fluid equations} \longrightarrow \text{sound speed and damping.}$$

In this section only, the standard sound-literature symbols $\eta = \mu$ and $\kappa = \lambda$ avoid confusing viscosity with chemical potential. Sound propagation uses a macroscopic closure informed by the computed transport coefficients. It is distinct from directly evolving the projected transverse kinetic equation of the preceding section.

3.2 An experimentally motivated kernel and units

For low-energy identical-boson scattering with scattering length a, the isotropic total cross section is $8\pi a^2$. In our full-solid-angle convention,

$$B = 2a^2 g, \qquad d^2 = 8a^2. \tag{42}$$

The same dimensionless Galerkin solver can therefore be used, with

$$\eta = \mu_{\text{num}} \frac{\sqrt{m k_{\mathrm{B}} T}}{8a^2}, \qquad \kappa = \lambda_{\text{num}} \frac{k_{\mathrm{B}} \sqrt{k_{\mathrm{B}} T/m}}{8a^2}. \tag{43}$$

This assumes the low-energy, energy-independent cross section remains an adequate description. Finite-energy scattering or many-body corrections would modify this model rather than the statistical sign alone.

The selected normal-phase data concern ^{39}K at approximately $1.3T_c$. We retain five observations with the published inverse Knudsen number $I = L/\ell \geq 3$, favouring the more hydrodynamic part of the dataset. The density is reconstructed using the published derived quantity:

$$n = \frac{I}{8\pi a^2 L}. \tag{44}$$

This makes the comparison conditional on that density reconstruction; it is not an independent determination of density from raw experimental images. One additional source row has inconsistent length conventions when its reported damping ratio is checked (the listed $70\,\mu$m corresponds to about $72\,\mu$m in that ratio). It is retained unchanged in the data archive and already lies outside the selected hydrodynamic subset.

3.3 Why the thermodynamic model must also change

The ideal-gas kinetic pressure alone is not the thermodynamic model used for this weakly interacting Bose experiment. In the normal Hartree–Fock approximation, introduce

$$g_{\rm int} = \frac{4\pi\hbar^2 a}{m}, \qquad z = \exp\!\left(\frac{\mu_{\rm chem} - 2g_{\rm int}n}{k_{\rm B}T}\right). \tag{45}$$

The effective single-particle energy is shifted by $2g_{\rm int}n$. The occupation still has the Bose form in terms of the effective fugacity z, while the total pressure and energy density acquire interaction terms:

$$p_{\rm kin} = nk_{\rm B}T\frac{G_{5/2}(z)}{G_{3/2}(z)}, \qquad p = p_{\rm kin} + g_{\rm int}n^2, \qquad \mathcal{U} = \frac{3}{2}p_{\rm kin} + g_{\rm int}n^2. \tag{46}$$

The kinetic transport source still contains $Z_{\rm kin}$, not $p/(nk_{\rm B}T)$ including the interaction pressure. To see why, the mean-field acceleration is $\mathbf{X} = -2g_{\rm int}\nabla n/m$. The Euler equation for the total pressure then gives

$$D_t\mathbf{u} - \mathbf{X} = -\frac{\nabla(p_{\rm kin} + g_{\rm int}n^2)}{mn} + \frac{2g_{\rm int}}{m}\nabla n = -\frac{\nabla p_{\rm kin}}{mn}.$$

This cancellation preserves the matched thermal source used in the collision calculation. Adding an interaction pressure without its corresponding force would be inconsistent.

Let $p_n = (\partial p/\partial n)_T$ and $p_T = (\partial p/\partial T)_n$. The derivatives used by the sound calculation are

$$p_n = k_{\rm B}T\frac{G_{3/2}}{G_{1/2}} + 2g_{\rm int}n, \tag{47}$$

$$p_T = nk_{\rm B}\left(\frac{5}{2}\frac{G_{5/2}}{G_{3/2}} - \frac{3}{2}\frac{G_{3/2}}{G_{1/2}}\right), \tag{48}$$

$$C_V = \frac{3}{2}p_T, \qquad C_P = C_V + \frac{Tp_T^2}{np_n}, \qquad c_{\rm ad}^2 = \frac{C_P}{C_V}\frac{p_n}{m}. \tag{49}$$

Here C_V and C_P are heat capacities per unit volume. Their units matter: they must not be mixed with heat capacities per unit mass or per particle.

3.4 Deriving the sound eigenproblem

Linearize continuity, longitudinal momentum and temperature about a uniform state. With $\rho = mn$ and zero bulk viscosity, they read

$$\partial_t\delta n = -n\partial_x u, \tag{50}$$

$$\rho\partial_t u = -\partial_x(p_n\delta n + p_T\delta T) + \frac{4}{3}\eta\partial_x^2 u, \tag{51}$$

$$C_V\partial_t\delta T = -Tp_T\partial_x u + \kappa\partial_x^2\delta T. \tag{52}$$

The term $4\eta/3$ is the longitudinal projection of the same traceless viscous stress derived earlier. Thermal conduction couples temperature to the density and velocity oscillation, so both computed coefficients contribute to sound damping.

Insert $e^{i(kx-\omega t)}$, take $k = \pi/L$ for the fundamental mode and introduce

$$C = \frac{\omega}{k c_{\mathrm{ad}}}, \qquad q_\eta = \frac{4\eta k}{3\rho c_{\mathrm{ad}}}, \qquad q_\kappa = \frac{\kappa k}{C_V c_{\mathrm{ad}}}. \tag{53}$$

Eliminating the three amplitudes gives

$$C^3 + i(q_\eta + q_\kappa)C^2 - (1 + q_\eta q_\kappa)C - i q_\kappa \frac{C_V}{C_P} = 0. \tag{54}$$

We select the damped propagating sound root. The quantities compared with the published data conventions are

$$c_I = \frac{|\omega|}{k}, \qquad \Gamma = -\operatorname{Im}\omega. \tag{55}$$

Thus c_I is not simply $\operatorname{Re}\omega/k$ when damping is appreciable. The dataset quotes $\Gamma/(2\pi)$ in hertz; the factor 2π must be retained when converting the eigenvalue. In the weak-damping limit,

$$\Gamma \simeq \frac{k^2}{2}\left[\frac{4\eta}{3\rho} + \kappa\left(\frac{1}{C_V} - \frac{1}{C_P}\right)\right]. \tag{56}$$

The code uses the full cubic rather than imposing this last asymptotic formula.

3.5 Results and the limits of this comparison

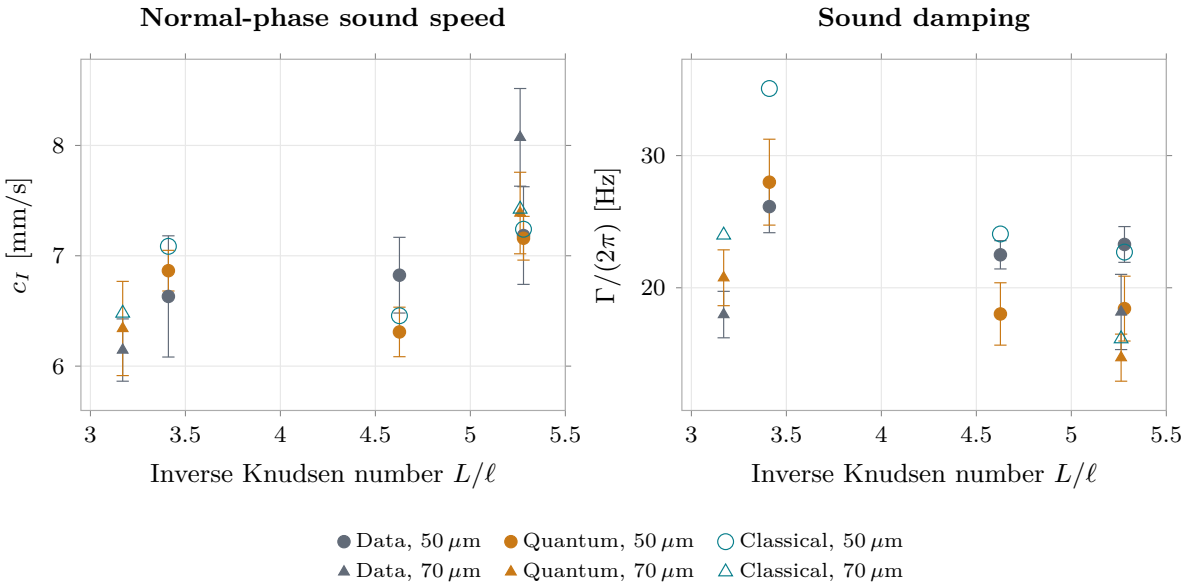


Figure 5: Five selected normal-phase observations from Hilker et al., plotted against inverse Knudsen number. Circle and triangle markers distinguish $L = 50$ and $70\,\mu$m boxes. Colors distinguish measurements, quantum transport and classical transport, the latter two using identical Hartree–Fock thermodynamics and linear Navier–Stokes–Fourier sound. Data bars are reported measurement uncertainties. Quantum-model bars are one-input-at-a-time sensitivity ranges, not confidence intervals; classical markers show central predictions only. No lines connect states with different physical inputs. Data: Ref. [4], CC BY 4.0.

			c_I [mm/s]		$\Gamma/(2\pi)$ [Hz]	
Index	L [μm]	a/a_{B}	Data	Model	Data	Model
1	50	670	6.632	6.865	26.14	27.99
2	50	740	6.824	6.310	22.50	18.02
3	50	800	7.183	7.159	23.28	18.42
4	70	800	6.146	6.341	17.97	20.76
5	70	1000	8.073	7.387	18.17	14.71

Table 5: Central values for the observations in Fig. 5. a_{B} is the Bohr radius. No parameters were fitted to these sound measurements.

For this extension we increase the basis to $N = 7$. Across the five states, the largest change from $N = 5$ to $N = 7$ is about 0.0351% in viscosity and 0.1679% in conductivity. Refining the collision quadrature from $(24, 16, 20)$ to $(32, 20, 24)$ changes these coefficients by less than 0.001%. These numerical changes are much smaller than the central-value differences from experiment: approximately -8.5% to $+3.5\%$ for c_I and -20.9% to $+15.5\%$ for damping.

Classical-transport control. We repeat the sound prediction with classical viscosity and conductivity at the same basis size and physical state, leaving the Hartree–Fock thermodynamics and cross-section convention unchanged. This isolates the transport correction; it is not a fully classical thermodynamic model. For either measured observable y, define the descriptive normalized RMS residual

$$R_y = \left[\frac{1}{5} \sum_{j=1}^{5} \left(\frac{y_j^{\mathrm{model}} - y_j^{\mathrm{data}}}{\sigma_{y,j}^{\mathrm{data}}} \right)^2 \right]^{1/2} .$$

The report exporter computes the following values from the stored predictions:

Model (same HF thermodynamics)	R_{c_I}	R_Γ
Quantum transport	1.031	2.654
Classical transport	1.040	2.648

The RMS agreement is nearly unchanged. Although the largest relative damping error decreases from 34.2% for classical transport to 20.9% for quantum transport, these five observations do not demonstrate an overall improvement from quantum transport corrections. Neither calculation fits the measured sound values. The residual normalization uses only the quoted measurement uncertainty: it does not include propagated input uncertainty or covariance and is not a formal significance test. The value $R_{c_I} = 1.031$ is an aggregate and does not mean that every point agrees within that many uncertainties.

The comparison therefore provides a useful application with a visible remaining discrepancy, rather than precision experimental validation of the collision solver. Possible limitations include the hydrodynamic closure at finite L/ℓ, the normal-phase thermodynamic approximation, the assumed cross section and the reconstruction of state variables. Their individual contributions have not been separated. Matching a separately published theory curve is another algebraic/unit check and should not be counted as matching the measured data.

The repository also evolves an illustrative standing wave initialized with a 1% density perturbation and zero initial velocity and temperature perturbation. This illustrates Eq. (52); it does not reproduce the experimental shaking and preparation protocol. The main result of the project remains the verified collision-to-transport computation. This final application shows how those coefficients enter an experimentally measurable quantity, and where further physical modelling becomes necessary.

References

[1] E. A. Uehling and G. E. Uhlenbeck, *Transport Phenomena in Einstein–Bose and Fermi–Dirac Gases. I*, Physical Review **43**, 552–561 (1933). doi:10.1103/PhysRev.43.552.

[2] L. Wu, *A fast spectral method for the Uehling–Uhlenbeck equation for quantum gas mixtures*, arXiv:1810.03090v1 (2018). Version used for the Eq. (50) benchmark.

[3] T. Hilker et al., *First and Second Sound in a Compressible 3D Bose Fluid*, Physical Review Letters **128**, 223601 (2022). doi:10.1103/PhysRevLett.128.223601.

[4] T. Hilker et al., *Research data supporting "First and Second Sound in a Compressible 3D Bose Fluid"*, University of Cambridge repository (2022), CC BY 4.0. doi:10.17863/CAM.83837.